

## UNIT-I & II

### 1. Check List for reporting in cyber crime at Cyber crime police Station

*Ans: Check list for Complainant*

**Please keep this information ready before filing your complaint:**

#### Mandatory Information

1. Incident Date and Time.
2. Incident details (minimum 200 characters) without any special characters (#\$%^&\*~|!).
3. Soft copy of any national Id ( Voter Id, Driving license, Passport, PAN Card, Aadhar Card) of complainant in .jpeg, .jpg, .png format (file size should not more than 5 MB).
4. In case of financial fraud, please keep following information ready:
  - i) Name of the Bank/ Wallet/Merchant
  - ii) 12-digit Transaction id/UTR No.
  - iii) Date of transaction
  - iv) Fraud amount
5. Soft copy of all the relevant evidences related to the cyber crime (not more than 10 MB each)

#### Optional/Desirable Information:

1. Suspected website URLs/ Social Media handles (wherever applicable)
2. Suspect Details (if available)
  - i) Mobile No
  - ii) Email id
  - iii) Bank Account No
  - iv) Address
  - v) Soft copy of photograph of suspect in .jpeg, .jpg, .png format (not more than 5 MB)
  - vi) Any other document through which suspect can be identified.

**CITIZEN LOGIN**

[Click Here for New User](#)

**LOGIN ID: \***

**MOBILE NO: \***

**OTP: \***

[Forgot Login Id](#)

## 2. Check list for Reporting Cyber Crime Online.

### How to file a Cyber Crime Complaint online?

The online portal where a victim can file a cyber crime complaint is <https://cybercrime.gov.in/Accept.aspx>, an initiative of Government of India that caters to complaints pertaining to the online Child Pornography (CP), Child Sexual Abuse Material (CSAM) or sexually explicit content and other cyber crimes such as social media crimes, online financial frauds, ransomware, hacking, crypto currency crimes, and online cyber trafficking. The portal also provides an option of reporting an anonymous complaint about reporting Child Pornography (CP) or sexually explicit content. One can follow below-mentioned steps to report a cyber crime online –

**STEP 1:** Go to [https://cyber](https://cybercrime.gov.in/Accept.aspx)

[crime.gov.in/Accept.aspx](https://cybercrime.gov.in/Accept.aspx) **STEP 2:** Click on 'Report

other cyber crimes' on the menu. **STEP 3:** Click on

'File a Complaint'.

**STEP 4:** Read the conditions and accept them.

**STEP 5:** Register your mobile number and fill in your name and State.

**STEP 6:** Fill in the relevant details about the offence.

## 3. Reporting phishing e-mails

### Avoid and report phishing emails

Learn how to spot deceptive requests online and take recommended steps to help protect your Gmail and Google Account.

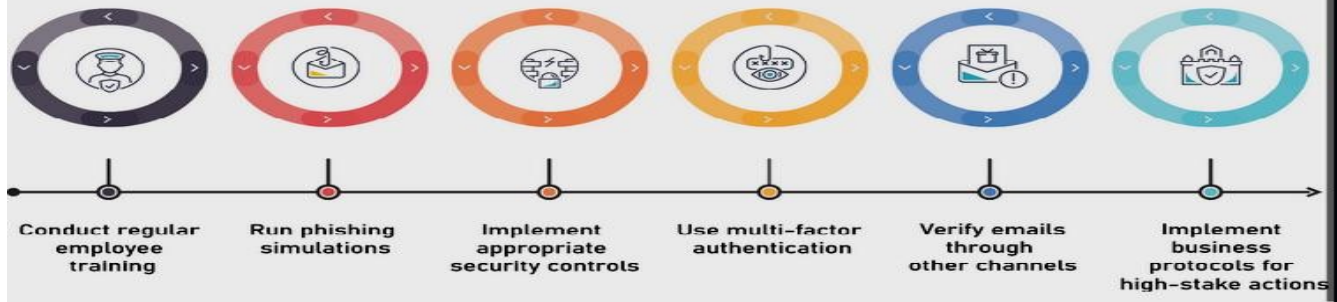
#### What phishing

Phishing is an attempt to steal personal information or break in to online accounts using deceptive emails, messages, ads, or sites that look similar to sites you already use. For example, a phishing email might look like it's from your bank and request private information about your bank account.

**Phishing messages or content may:**

- Ask for your personal or financial information.
- Ask you to click links or download software.
- Impersonate a reputable organization, like your bank, a social media site you use, or your workplace.
- Impersonate someone you know, like a family member, friend, or coworker.
- Look exactly like a message from an organization or person you trust.

## BEST PRACTICES TO PREVENT PHISHING EMAIL ATTACKS



#### 4. **Demonstration of email phishing attack and preventive measures.**

Internet pirates steal personal financial information with a new type of Internet piracy called phishing, pronounced "fishing," and that's exactly what these thieves are doing: "fishing" for your personal financial information.

What they want are account numbers, passwords, Social Security numbers, and other confidential information that they can use to loot your checking account or run up bills on your credit cards. In the worst case, you could find yourself a victim of identity theft. With the sensitive information obtained from a successful phishing scam, these thieves can take out loans or obtain credit cards and even driver's licenses in your name. They can do damage to your financial history and personal reputation that can take years to unravel.

But if you understand how phishing works and how to protect yourself, you can help stop this crime.

### How to Protect Yourself

1. **Never provide your personal information in response to an unsolicited request,** whether it is over the phone or over the Internet. Emails and Internet pages created by phishes may look exactly like the real thing. They may even have a fake padlock icon that ordinarily is used to denote a secure site. If you did not initiate the communication, you should not provide any information.
2. **If you believe the contact may be legitimate, contact the financial institution yourself.** You can find phone numbers and Websites on the monthly statements you receive from your financial institution, or you can look the company up in a phone book or on the Internet. The key is that you should be the one to initiate the contact, using contact information that you have verified yourself.
3. **Never provide your password over the phone or in response to an unsolicited Internet request.** A financial institution would never ask you to verify your account information online. Thieves armed with this information and your account number can help themselves to your savings.
4. **Review account statements regularly to ensure all charges are correct.** If your account statement is late in arriving, call your financial institution to find out why. If your financial

institution offers electronic account access, periodically review activity online to catch suspicious activity.

## UNIT- III

### 5. Basic Check list, privacy and security settings of popular Social media Platforms.

#### Social Media Security Checklist

Keep your personal data safe while using social media

##### Manage Privacy Settings

These settings allow you to make sure you are only sharing with friends and family. These settings can change with updates so you should check them regularly.

##### Don't Enable Auto Login

If someone gets access to your phone or computer make sure they can't automatically access your social sites. Make sure your computer's browser "remember" your login and password isn't set.

##### Use Two-factor authentication 2FA

'Double checking' you really are who you say you are by texting your phone a code that you need to enter before you can log in.

##### Setup Security Answers

You can set up security questions on your accounts use questions that are hard to find out about you.

##### Use a VPN

Virtual Private Network is an encryption tool. If you want to keep your conversations, messages, and calls secure this is the best solution.

##### Keep Antivirus Updated

Never go on the internet without installing anti-malware and antivirus software to secure your online activity and important data.

##### Change Passwords Frequently

Always use a combination of letters, numbers, and symbols to make it harder to break. Good, secure passwords shouldn't be easy to remember.

Never use the same password for multiple websites because they will all become compromised.

##### Selective Accepting Friends

Are they really who they say they are? Do you really know that their profile is real and not fake? The safest option is to only accept "friend" requests from people you know in the real world.

### **Turn Off GPS**

Turn off your GPS to keep your exact location private. If you plan to share images online turn off the GPS function to protect your safety.

### **Internet is Permanent**

Always consider that anything you post stays there forever. Even if you delete your account, someone may already print/copy your content.

### **Personal Information**

Careful who you reveal personal information. Always be suspicious of anyone who asks for your personal information over any social media platform.

### **Beware Clicking on Links**

Even if sent from your friends think twice. Hackers love social platforms because you are more likely to click on something from your friends.

### **Block Fake Accounts**

Sadly social media is full of fake profiles. These can be used for a number of reasons from hacking or monitoring your activities.

### **Suspicious Login Attempts**

Social media platforms email you if they feel a login is suspicious. Maybe because it comes from a different IP address, a different browser or your account is sending spamming posts.

Keep checking your emails regularly. If you get a suspicious login attempt to your account, change your password as soon as possible.

### **Keep Browser Updated**

Make sure you have the latest version of your browser that is not vulnerable to attacks and hackers.

Don't save passwords in your browser because if your system gets compromised your passwords can be easily read.

### **Auto Share Options**

There are options to automatically share posts from one platform to another.

### **Close Old Accounts**

Old social media accounts you haven't used in years can risk your personal data security. Delete account and remove as much personal information as possible.

### **App Authorization**

Every time you log in to another site using the option of using a social media login you are granting authorisation to your account.

They tell you what data this app will have access to before you agree but if you stop using that site remember to revoke authorisation.

## Password Manager

Your passwords are valuable, so treat them that way.

There is a number of password manager software available that help you store and organise passwords. This enables you to log in with just one main password that then allows you to access all your accounts.

## 6. Reporting and redressal mechanism for violations and misuse of Social media Platforms.

Some common issues reported on social media:

- **Reporting abusive content:** Users can report content that they find abusive, offensive, or in violation of platform policies. These reports are reviewed by the help centre.
- **Reporting CSAM (Child Sexual Abuse Material):** CSAM content can be reported to platform help centre. Social media platforms have stringent policies in place to address such concerns and ensure a safe digital environment for everyone, including children.
- **Reporting Misinformation or Fake News:** With the proliferation of misinformation online, users can report content that they find or suspect misleading or false information and Fact-checking bodies are employed to assess the accuracy of reported content.
- **Content violating intellectual property rights:** If there is a violation or infringement of any intellectual property work, it can be reported on the platform.
- **Violence of commercial policies:** Products listed on social media platforms are also needed to comply with the platform's Commercial Policies.

### Background:

The Digital India programme has now become a movement which is empowering common Indians with the power of technology. The extensive spread of mobile phones, Internet etc. has also enabled many social media platforms to expand their footprints in India. Common people are also using these platforms in a very significant way. Some portals, which publish analysis about social media platforms and which have not been disputed, have reported the following numbers as user base of major social media platforms in India:

- WhatsApp users: 53 Crore
- YouTube users: 44.8 Crore
- Facebook users: 41 Crore
- Instagram users: 21 Crore
- Twitter users: 1.75 Crore

## • UNIT- IV

## 7. Configuring Security settings in Mobile Wallets and UPIs.

Choose a Strong Password or PIN:

- Set a strong and unique password or personal identification number (pin) for your mobile wallet and UPI application.



- Avoid using common passwords or easily guessable combinations like birth dates or phonenumber.

#### **Enable Biometric Authentication:**

- If your mobile device supports biometric authentication methods such as fingerprints or facial recognition, enable these options for added security.
- Biometric provide an extra layer of protection in addition to your password or pin.

#### **Keep your mobile device updated:**

- Regularly update your mobile devices operating system and keep your mobile wallet and UPI application up to date.
- Software updates often include security patches that address vulnerabilities.

#### **Enable Two –Factor Authentication (2FA):**

- Whenever possible, enable two factor authentications for your mobile wallet and UPI application.
- This typically involves entering a verification code sent to your registered mobile number or email. Address, providing an extra layer of security during login or transaction confirmation.

#### **Use secure Networks:**

- Avoid using public wifi network or unsecured internet connections while conducting financial transactions.
- Use a trusted network or enable your mobile data connection for secure communication.

### **8. Check list secure net**

#### **banking Online Banking**

#### **Checklist**

1. **Protect and memorize your passwords.** Never write your passwords down or share them with anyone. Change passwords regularly and use combinations of letters, numbers and "special characters" such as the #, @, ), ? Signs. Do not use your Social Security Number or birthday as a username or password. Never use your username as a password.
2. **Keep your computer operating system up to date.** If your computer is older than five years, its operating system may not offer the same level of protection as newer systems. System manufacturers provide frequent updates to help make your system more secure. Regularly check for updates from your system manufacturer.
3. **Use a current web browser.** Cathay Bank continually upgrades our online services to provide you with the most secure online services. We do not allow access to Cathay Online Banking or Cathay Business Online Banking using browsers that do not meet our security criteria. If you have problems accessing our online banking services, you may need to upgrade your browser.
4. **Install a personal firewall.** Firewalls provide additional protection for your home computer. Many operating systems come with firewall protection. Before



purchasing a firewall, check your operating system.

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5. **Antivirus and Antispyware Software.** Virus and Spyware protection software helps reduce the risk of contracting computer malwares (viruses and spyware) that can compromise your computer and personal security while online.
6. **Use secure websites for transactions and shopping.** Make sure the web page you are viewing offers encryption of your data. If you see a lock symbol in the lower right-hand corner of your browser window, or if the web address of the page you are viewing begins with https://, this indicates that web page is secure and uses encryption. Cathay Bank provides 2048-bit encryption, the highest level available today.
7. **Do not open e-mail or download programs from unknown sources.** Sometimes hidden programs or viruses are contained in e-mail or on downloaded programs which can compromise your computer. Use caution when downloading from an unfamiliar site.
8. **Terminate the Internet when not in use.** Take an additional precaution by terminating your Internet session when not in use. This will help to avoid unwanted access to your computer and its data.

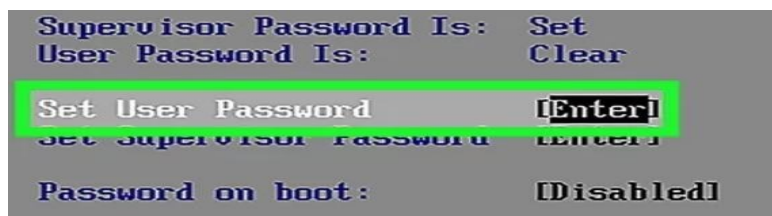
## UNIT- V

### 9. Setting, configuring and managing three password policy in the computer (BIOS, Administrator and Standard User)

#### Set a BIOS Password



1. **Protect your laptop with BIOS passwords.** A BIOS password is the extremely strong password that locks up the hardware and makes the laptop completely unusable. Only logging with the password can you enter into the operating system.



2. **Create a BIOS password.** Restart the laptop, and press F2 continuously to the following interface. Select the security with the cursor and choose "Set User" password or "Set UserPassword".

Note: The difference between Set User Password and Set Supervisor Password: User password controls access to the system at boot; supervisor password controls access to the setup utility.



3. Press Enter, and fill the three blanks with your password.
  4. Press Enter and pop up Setup Notice, which means that you have reset BIOS password.
  5. Press F10 to save it and select Yes to exit, your laptop will log on automatically.
  6. **Ensure a reset.** It is challenging if you forget a BIOS password. Reset the BIOS password using standard BIOS backdoor passwords to reset BIOS password. A backdoor is one of the best ways to protect BIOS password which is provided for users to access the BIOS when the hardware is being maintained. Note that some types of backdoor passwords will cease to work when entering the wrong password for more than three times. Here are some well-known backdoor passwords:
  7. **Protect your laptop with Windows Password.** A Windows password is the confirmation code that a PC user requesting access to the Windows system is really that particular user.
    - Create an administrator password with which the owner can log on in to the Windows system. And then you can create a Windows password reset disk.
10. **Setting and configuring two factor authentications in the Mobile phone.**

#### Set up your mobile device to use a text message as your verification method

1. On the **Additional security verification** page, select **Authentication phone** from the **Step 1: How should we contact you** area, select your country or region from the drop-down list, and then type your mobile device phone number.
2. Select **Send me a code by text message** from the Method area, and then select **Next**.
3. Type the verification code from the text message sent from Microsoft into the **Step 2: We've sent a text message to your phone** area, and then select **Verify**.
4. From the **Step 3: Keep using your existing applications** area, copy the provided app password and paste it somewhere safe.

5. Select **Done**.

### Set up your mobile device to receive a phone call

1. On the **Additional security verification** page, select **Authentication phone** from the **Step 1:How should we contact you** area, select your country or region from the drop-down list, and then type your mobile device phone number.
2. Select **Call me** from the **Method** area, and then select **Next**.
3. You'll receive a phone call from Microsoft, asking you press the hash tag (#) sign on your mobile device to verify your identity.
4. From the **Step 3: Keep using your existing applications** area, copy the provided app password and paste it somewhere safe.

## 11. Security Patch management and updates in computer and Mobiles.

Patch management creates a centralized process for applying new patches to IT assets. These patches can improve security, enhance performance, and boost productivity.

### Security updates

Security patches address specific [security risks](#), often by remediating a particular vulnerability.

Hackers often target unpatched assets, so the failure to apply security updates can expose a company to security breaches.

### Feature updates

Some patches bring new features to apps and devices. These updates can improve asset performance and user productivity.

### Bug fixes

Bug fixes address minor issues in hardware or software. Typically, these issues don't cause security problems but do affect asset performance.

### Minimizing downtime

Most companies find it impractical to download and apply every patch for every asset as soon as it's available. That's because patching requires downtime. Users must stop work, log out, and reboot key systems to apply patches.

### Regulatory compliance

Under regulations like the General Data Protection Regulation (GDPR), the Health Insurance Portability and Accountability Act (HIPAA), and the Payment Card Industry Data Security Standard (PCI-DSS), companies must follow certain [cybersecurity](#) practices. Patch management can help organizations keep critical systems compliant with these mandates.

## 12. Managing Application permissions in Mobile phone.

### Change app permissions

1. On your device, open the Settings app.
2. Tap **Apps**.
3. Tap the app you want to change. If you can't find it, tap **See all apps**. Then, choose your app.
4. Tap **Permissions**.
  - o If you allowed or denied any permission for the app, you'll find them here.
5. To change a permission setting, tap it, then choose **Allow** or **don't allow**.

For location, camera, and microphone permissions, you may be able to choose:

- **All the time:** For location only. The app can use the permission at any time, even when you're not using the app.
- **Allow only while using the app:** The app can use the permission only when you're using that app.
- **Ask every time:** Every time you open the app, it'll ask to use the permission. It can use the permission until you're done with the app.
- **Don't allow:** The app cannot use the permission, even when you're using the app.

### Change permissions based on their type

You can check which apps have the same permission setting. For example, you can check which apps have permission to check your calendar.

1. On your device, open the Settings app.
2. Tap **Security & Privacy** > **Privacy** > **Permission manager**.
3. Tap a permission type.
  - o If you allowed or denied permission to any apps, you'll find them here.
4. To change an app's permission, tap the app, and then choose your permission settings.

## 13. Installation and configuration of computer Anti-virus.

### What is Antivirus?

A computer antivirus is a software tool that is developed to detect, remove, or prevent computer viruses and other malware from a computer and similar devices. This software kind usually spends some time scanning for any patterns or behaviour that may be done by a malicious code like a virus, worm, Trojan, or ransomware.

## Why do we Need Antivirus?

Antivirus protection is the program helping us to fulfil the aim of keeping our devices safe from viruses, malware, and [ransomware](#). Such harmful software can be very dangerous, it can just infect our computers and steal our personal information, corrupt files, or completely devastate a designated device and make it unusable.

## How to Install Antivirus on Computer Step by Step?

### Step 1: Download a Reliable Antivirus Software

The first thing to do when you want to install antivirus software on your Windows 10 OS is you have to download dependable [antivirus software](#).

### Step 2: Install the Antivirus Software

When you have downloaded the antivirus software, click the 'Open a file' button and after that click the 'Install' button. Installation is convenient.

### Step 3: Configure the Software Settings

When you company the antivirus software, then you can set the settings to give you the best protection according to you.

### Step 4: Start the Scan

When the antivirus software has been installed, properly configured, and activated, you can start the scan. The scan will detect all the problems on your PC and delete them from your system effortlessly.

### Step 5: Resolve the issues

After the scan, all issues or viruses that have been found in your computer are displayed. Click on Resolve all issues to remove the virus from pc.

### Step 6: Monitor the Software Performance

When you are done installing and setting up the antivirus app, you need to follow up on its performance.

## Configuring Anti-Virus

This section describes how to configure Anti-Virus.

### Preparing

Before enabling Anti-Virus, make the following preparations:

1. Make sure your system version supports Anti-Virus.
2. Import an Anti-Virus license and reboot. The Anti-Virus will be enabled after the rebooting.

### Creating an Anti-Virus Rule

To create an Anti-Virus rule, take the following steps:

1. On the Navigation pane, click **Configure > Security > Anti-Virus** to visit the Anti-Virus page.
2. Click **New**.
3. In the Anti-Virus Rule Configuration dialog, type the name into the **Rule name** box.
4. After the **Binding zone** option, select a security zone for the Anti-Virus rule from the **Available** list, and click  to add to the **selected** list.
5. Specify a protection type. If **Predefined** is selected, you need to specify a security level for the protection in the Security level section. The default level is Medium. If **User-defined** is selected, you need to specify the file types and protocol types, as well as the actions that will be taken if any virus has been detected

## 14. Installation and Configuration of computer Host Firewall.

A firewall plays a vital role in [network security](#) and needs to be properly configured to keep organizations protected from data leakage and cyber attacks.

This is possible by configuring domain names and Internet Protocol (IP) addresses to keep the firewall secure. Firewall policy configuration is based on network type, such as public or private, and can be set up with security rules that block or allow access to prevent potential attacks from hackers or malware.

Proper [firewall configuration](#) is essential, as default features may not provide maximum protection against a cyberattack.

### How To Configure a Firewall

#### 1. Secure the firewall

Securing a firewall is the vital first step to ensure only authorized administrators have access to it. This includes actions such as:

1. Update with the latest firmware
2. Never putting firewalls into production without appropriate configurations in place
3. Deleting, disabling, or renaming default accounts and changing default passwords
4. Use unique, secure passwords
5. Never using shared user accounts. If a firewall will be managed by multiple administrators, additional admin accounts must have limited privileges based on individual responsibilities
6. Disabling the [Simple Network Management Protocol \(SNMP\)](#), which collects and organizes information about devices on IP networks, or configuring it for secure usage
7. Restricting outgoing and incoming network traffic for specific applications or the [Transmission Control Protocol \(TCP\)](#)

#### 2. Establish firewall zones and an IP address structure

It is important to identify network assets and resources that must be protected. This includes creating a structure that groups corporate assets into zones based on similar functions and the level of risk.

#### 3. Configure access control lists (ACLs)

Access control lists (ACLs) enable organizations to determine which traffic is allowed to flow in and out of each zone. ACLs act as firewall rules, which organizations can apply to each firewall interface and sub interface.

#### 4. Configure other firewall services and logging

Some firewalls can be configured to support other services, such as a [Dynamic Host Configuration Protocol \(DHCP\)](#) server, [intrusion prevention system \(IPS\)](#), and Network Time Protocol (NTP) server. It is important to also disable the extra services that will not be used.

#### 5. Test the firewall configuration

With the configurations made, it is critical to test them to ensure the correct traffic is being blocked and that the firewall performs as intended.

## **6. Manage firewall continually**

Firewall management and monitoring are critical to ensuring that the firewall continues to function as intended. This includes monitoring logs, performing vulnerability scans, and regularly reviewing rules.

## **15. Wi-Fi Security management in computer and Mobile.**

Mobile devices are very subjective tools, mainly used to facilitate routine operations, but they also serve to store very confidential personal knowledge and data. Mobile security is the safety and protection of smartphones, tablets, laptops and other portable devices, and the systems they connect to, from threats and vulnerabilities linked with wireless computing.

**SMiShing:** SMiShing attack is initiated via text messages as an alternative of email.

**BYOD:** As enterprise users are permitted advanced admittance/access from individual mobiles gadgets like smartphones and tablets are efficiently substituting desktops for numerous business duties.

**The Internet of Things (IoT):** With the numerous kinds of smart devices, they cannot constantly be supervised by customers or antivirus software's. This makes IoT systems an attractive objective for invaders who make use of them as entry points to the bigger network.

- **Attacks based on GSM networks**

Once the encryption algorithm of GSM is cracked, the invader can capture all unencrypted messages made by the target's/user's Smartphone system.

- **Attacks based on lack of Wi-Fi security**

An invader can attempt to snoop on Wi-Fi communications to obtain intelligence (e.g username, password).

- **Attacks based on Bluetooth**

A mobile phone must be in the reach and Bluetooth in discovery mode. The invader sends a file/script via Bluetooth. If the receiver acknowledges the request, a virus is broadcast.

Example: Cabir is a worm that circulates via Bluetooth link/connection.

- **Attacks based on Operating system**

Occasionally it is feasible to conquer the security defences by altering the OS itself.

